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Effects of Exercise Sequence and Velocity Loss Threshold During Resistance Training on Following Endurance and Strength Performance During Concurrent Training

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Purpose: This study aimed to analyze the response to 4 concurrent training interventions differing in the training sequence and in the velocity loss (VL) threshold during strength training (20% vs 40%) on following endurance and strength performance. ***Methods:*** A randomized crossover research design was used. Sixteen trained men performed 4 training interventions consisting of endurance training (ET) followed by resistance training (RT), with 20% and 40% VL, respectively (ET + RT20 and ET + RT40), and RT with 20% and 40% VL, respectively, followed by ET (RT20 + ET and RT40 + ET). The ET consisted of running for 10 minutes at 90% of maximal aerobic velocity. The RT consisted of 3 squat sets with 60% of 1-repetition maximum. A 5-minute rest was given between exercises. The oxygen uptake throughout the ET and repetition velocity during RT were recorded. The blood lactate concentration, vertical jump, and squat velocity were measured at preexercise and after the endurance and strength exercises. ***Results:*** The RT40 + ET protocol showed an impaired running time along with higher ventilatory equivalents compared with those protocols that performed the ET without previous fatigue. No significant differences were observed in the repetitions per set performed for a given VL threshold, regardless of the exercise sequence. The protocols consisting of 40%VL induced greater reductions in jump height and squat velocity, along with elevated blood lactate concentration. ***Conclusions:*** A high VL magnitude (40%VL) induced higher metabolic and mechanical stress, as well as greater residual fatigue, on the following ET performance.

Keywords: velocity-based training, training volume, endurance training, maximal oxygen uptake, muscle fatigue, maximal aerobic speed

Numerous athletic disciplines require the development of strength and endurance performance concomitantly. Under such demands, endurance (ET) and resistance training (RT) are undertaken simultaneously within a training program, which is termed concurrent training (CT).[1] Most CT studies have shown that RT combined with ET induces greater gains in endurance performance than ET alone.[2,3] However, endurance and strength training can also interfere with each other,[4] resulting in attenuated strength gains compared with RT alone (ie, interference effect).[1,5] Accordingly, one of the main problems faced by strength and conditioning coaches is the question of how to manipulate the acute training variables (ie, exercise sequence, intensity, volume, and exercise mode) during CT to maximize both endurance and strength performance without interfering with one another.

Due to time restrictions and sport demands, some athletes are forced to perform both strength and endurance exercises within the same training session or in close proximity. Accordingly, the sequence of RT and ET within the session may be paramount for maximizing endurance and strength development.[6] In this regard, residual fatigue induced by the preceding endurance session may compromise the quality in the following resistance session.[7] Indeed, a reduction in the number of repetitions performed in RT immediately following an endurance session has been observed,[8] even up to 8 hours after an endurance session.[9] By contrast, it has also been suggested that muscle damage and residual fatigue induced by an RT session may acutely affect endurance performance,[10] which may induce suboptimal endurance development and performance.[6] Several studies have reported decreased running economy[11,12] and impaired endurance time-trial performances following a single RT set.[13–15] In addition, an attenuated endurance running performance has been observed up to 24 hours following strength training.[16] Therefore, as residual fatigue induced by a previous exercise bout may compromise performance in the subsequent exercise bout, the exercise sequence and the extent of the fatigue induced by each CT session appear critical.

The progressive impairment in lifting velocity across repetitions during RT provides a simple and objective means of quantifying the levels of fatigue incurred during a training set.[17] As a result, a training set should be stopped as soon as a prescribed velocity loss (VL) threshold is detected.[18,19] This approach is based on the relationships ($R^2 \geq .83$) reported between the VL magnitude and the level of fatigue,[17] along with the strong relationships ($R^2 = .93–.97$) observed between the VL magnitude and the percentage of repetitions that can be completed before muscle failure.[20] A VL of about 40% in the set means that the set is conducted close to muscle failure, whereas a VL of 20% means that the athlete has performed ~50% of the possible repetitions for the full-squat (SQ) exercise.[20] Thus, monitoring the VL magnitude during RT provides real-time feedback on the level of fatigue developed in the set, which may be useful in protocols aimed at applying more homogeneous stimuli across individuals, as a standardized state

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of fatigue is induced. Accordingly, it would be expected that the VL magnitude incurred within the RT sets influences the residual fatigue induced during CT. However, to our knowledge, no previous study has analyzed the effect of different VL magnitudes during CT on strength and endurance performance. As inappropriate scheduling of concurrent endurance and strength training has previously been implicated in attenuated strength improvement,[21,22] it appears that the manipulation of training variables such as exercise sequence and residual fatigue is critical to avoid potential interferences in CT settings. Hence, in an attempt to gain greater understanding of the acute responses to exercise sequence and the level of fatigue induced within the RT set during concurrent endurance and strength training, we aimed to analyze the mechanical and physiological response to 4 experimental protocols differing in (1) training sequence and (2) the VL magnitude allowed during the RT set.

Methods

Subjects

A total of 16 resistance- and endurance-trained men (age = 36.4 [10.0] y, body mass = 77.1 [7.6] kg, height = 176.3 [6.7] cm, 1-repetition maximum [1RM] in SQ = 105.2 [25.8] kg, maximal oxygen uptake [VO_2max] = 53.9 [5.2] mL·kg·min^{-1}, and velocity associated at VO_2max [vVO_2max] = 15.6 [1.4] km·h^{-1}) volunteered to participate in this study. The subjects had a CT background ranging from 2 to 5 years and regularly participated in 2 to 5 sessions per week (2–3 RT sessions plus 1–2 ET sessions). After being informed about the purpose, testing procedures, and potential risks of the investigation, the subjects gave their written consent to participate. The subjects also were screened for cardiovascular disorders through echocardiogram and electrocardiographic records during the initial endurance assessment. None of the subjects was taking drugs, medications, or dietary supplements known to influence physical performance. The study was approved by the Research Ethics Committee of Pablo de Olavide University and was conducted in accordance with the Declaration of Helsinki II.

Design

A randomized crossover research design was used to examine the effects of (1) the training sequence during CT: ET + RT versus RT + ET and (2) the magnitude of the VL induced during the RT set: 20% versus 40%. Therefore, 4 experimental protocols were randomly performed, as follows: endurance exercise followed by resistance exercise with 20% and 40% VL, respectively (ET + RT20 and ET + RT40); and resistance exercise with 20% and 40% VL, respectively, followed by endurance exercise (RT20 + ET and RT40 + ET). Each protocol was performed once per week. In order to examine the effects of different protocols on endurance and strength-training performance, the oxygen uptake (VO_2) attained throughout the endurance exercise and the velocity attained in each repetition during the resistance exercise were recorded. In addition, the blood lactate concentration, vertical countermovement jump (CMJ) height, and velocity attained against the absolute load (in kilograms) that elicited 1.00 m·s^{-1} in the baseline measure (V_1-load) were assessed at preexercise and after the endurance and strength exercises (Figure 1). These metabolic and mechanical measurements have previously been used to monitor fatigue induced by both endurance and strength exercises.[17,23] The V_1-load was chosen because it represents a sufficiently moderate loading intensity (~60% 1RM in SQ)[24] to allow detection of the effect of fatigue on movement velocity and it is quick to establish as part of the warm-up.[17]

The subjects underwent 2 familiarization sessions 2 weeks before the start of the first trial. During these sessions, the subjects performed an incremental SQ test until the attained mean propulsive velocity (MPV) was lower than 1.00 m·s^{-1} and several CMJ practices, while researchers emphasized proper technique. Initial endurance and strength assessments were performed on 2 separate days, 1 week before the first trial. Assessment of vVO_2max, via an incremental treadmill protocol, and 1RM strength accompanied by the individual load–velocity relationships were performed for the purpose of prescribing relative intensity in both endurance and resistance exercises. All protocols were performed at the same time of the day for each subject and under controlled environmental conditions (20°C–22°C and 55%–65% humidity) in a research laboratory.

Procedures

Initial Endurance Assessment. This test was performed on a treadmill (h/p/cosmos Quasar, Nussdorf-Traunstein, Germany) following the incremental treadmill protocol proposed by Léger and Boucher.[25] The running velocity began at 8 km·h^{-1} and was increased by 1 km·h^{-1} every 2 minutes until volitional exhaustion. The treadmill incline was maintained at a constant 1% throughout the test. Heart rate (HR) was recorded continuously using an HR sensor H2 (Polar Electro Oy, Kempele, Finland), and VO_2 was measured breath-by-breath throughout the test using a gas analyzer (Oxycon Mobile®; Jaeger, Hoechberg, Germany) that was calibrated before each test, using instructions provided by the manufacturer. The VO_2max value was defined as the highest 30-second average VO_2 during the test and was considered to have been reached if there was no increase (<100 mL·min^{-1}) in VO_2 with an increase in exercise intensity.[26] To ensure that VO_2max was reached, each subject had to meet the following criteria: respiratory exchange ratio > 1.00 and HR within 5% of their age-predicted maximum. The vVO_2max was defined as the minimum velocity at which the VO_2max was reached.

Initial Strength Assessment. The SQ was performed on a Smith machine (Multipower Fitness Line; Peroga Fitness, Murcia, Spain). A detailed description of SQ technique has been provided elsewhere[27] and was exactly reproduced throughout the study. The subjects were required to always execute the concentric phase in an explosive manner, at maximal intended velocity. The initial load was set at 30 kg and progressively increased in 10-kg increments until the attained MPV was <0.70 m·s^{-1}. Thereafter, the load was individually adjusted in smaller increments (5 down to 2.5 kg) until the MPV was <0.50 m·s^{-1} (ie, ≥90% 1RM). All velocity measures reported in this study refer to the MPV, which corresponds to the portion of the concentric action during which the measured acceleration is greater than the acceleration due to gravity (−9.81 m·s^{-2}).[28] Three repetitions were executed for light (>1.00 m·s^{-1}), 2 for medium (1.00–0.80 m·s^{-1}), and only 1 for the heaviest loads (<0.80 m·s^{-1}). Interset rests were 3 minutes for the light and medium loads and 5 minutes for the heaviest loads. Only the best repetition (ie, fastest MPV) for each load was considered for subsequent analysis. Owing to the very close relationship (R^2 = .95) between %1RM and MPV in the SQ exercise, the 1RM was estimated from the MPV attained against the heaviest load of the test (ie, ≥90% 1RM), as follows: %1RM = −5.961 MPV^2 −50.71 MPV + 117.0 (standard error of estimate = 4.0% 1RM).[24] A linear velocity transducer (T-Force System; Ergotech, Murcia, Spain) was

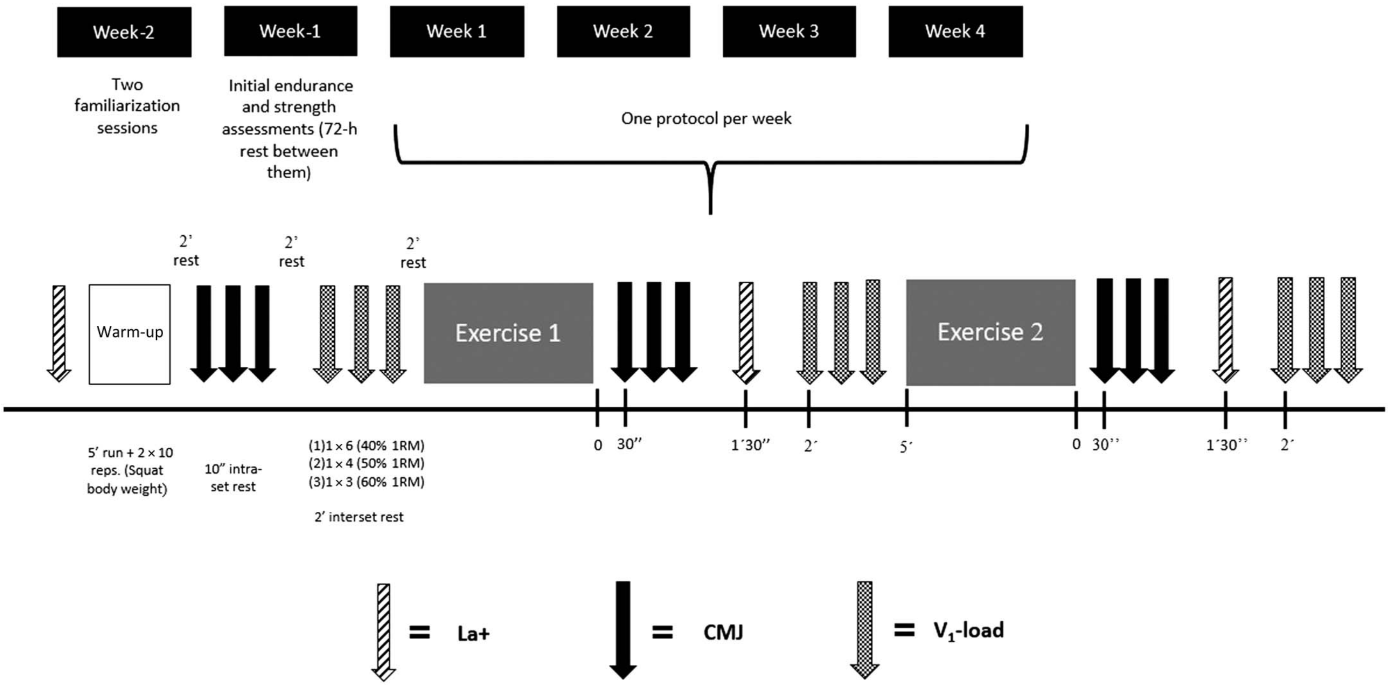

Figure 1 — Schematic representation of the study design. Mechanical and metabolic measurements were taken at different times during the session to analyze the fatigue induced by each exercise mode. Two protocols consisted of a first endurance exercise (10 min at ~90% vVO_2max) and a second resistance exercise (~60% 1RM with a loss of velocity of ~20% or ~40% in the deep squat exercise), and in the other 2 protocols, the resistance exercise preceded the endurance exercise. 1RM indicate 1-repetition maximum; La+, blood lactate; CMJ, countermovement jump; V_1-load, velocity attained against the absolute load that elicits 1 $m \cdot s^{-1}$ in the baseline measure (Pre).

used to measure the bar velocity. The reliability of this setup has been documented elsewhere.[17]

Concurrent Training Protocols. Four CT protocols were conducted (ET + RT20, ET + RT40, RT20 + ET, and RT40 + ET). The endurance exercise consisted of running on a treadmill for 10 minutes at 90% of the vVO_2max previously determined for each subject. The 2 RT sessions consisted of 3 sets with 60% 1RM in the SQ exercise, with 2-minute interset rests, but differed in the VL threshold allowed in each set (20% vs 40%). When the corresponding target VL limit was exceeded, the set was finished. A 5-minute rest period was given between the endurance and resistance exercises, or vice versa. To determine whether the protocol performed affected strength training performance, the velocity attained in each repetition was recorded using a linear velocity transducer (T-Force System; Ergotech). The repetitions performed in each set (repetition per set), the highest velocity measured in the 3 sets (Fastest-V), the slowest velocity measured in the 3 sets (Slowest-V), and the mean velocity of all repetitions during the 3 sets (Mean-V) were analyzed in order to compare the effects on RT performance.

To examine the effects on ET performance, a set of cardiovascular and respiratory variables, including HR, VO_2, ventilation volume, ventilatory equivalents for oxygen (VE/VO_2) and carbon dioxide (VE/VCO_2), and respiratory exchange ratio (ie, VCO_2/VO_2) were measured during the endurance exercise.

Measurements of Mechanical and Metabolic Response. Figure 1 provides a detailed description of the experimental protocol. The blood lactate concentration was measured before the warm-up. About 5 µL of whole blood from the finger was used for the quantification of lactate, which was measured with a portable lactate analyzer (Lactate Pro 2; Arkray, Kyoto, Japan). A standardized warm-up was performed, which consisted of the following: (1) 5-minute treadmill running at 8 $km \cdot h^{-1}$, (2) 2 sets of 10 squats with no external load, and (3) 2 sets of 5 and 3 CMJ of increasing intensity. This was followed by (4) 3 maximal CMJ, separated by 10-second rests, from which the mean jump height was taken as the preexercise reference value. An infrared timing system (OptojumpNext; Microgate, Bolzano, Italy) was used to determine the CMJ height. (5) Following the CMJ measurement, the V_1-load in SQ was determined. For this purpose, 6 repetitions with 40% 1RM (~1.28 [0.03] $m \cdot s^{-1}$), 4 repetitions with 50% 1RM (~1.14 [0.03] $m \cdot s^{-1}$), and 3 repetitions with 60% 1RM (V_1-load, ~1.00 [0.03] $m \cdot s^{-1}$) (2-min interset rests) were performed. The mean velocity of the 3 maximal intended repetitions with the V_1-load was registered as the preexercise reference value for this variable. (6) After 2-minute rests, the athletes performed the first exercise (endurance or resistance exercise). (7) Then, 30 seconds after completing exercise 1, the participants again performed 3 maximal CMJ, separated by 10-second rests. (8) In addition, 1 minute 30 seconds after completing exercise 1, the blood lactate concentration was measured, and (9) finally, another 3 repetitions with the V_1-load were performed (2 min after exercise 1). This sequence was strictly repeated after exercise 2. (10) Five minutes after the first exercise, the second exercise (endurance or resistance exercise) was performed. Strong verbal encouragement and velocity feedback were provided in every repetition throughout all exercise sets.

Statistical Analyses. The values are reported as mean (SD). Statistical significance was established at $P < .05$. The distribution of each variable was examined using the Shapiro–Wilk normality test. To examine whether the protocol (ET + RT20 vs ET + RT40 vs RT20 + ET vs RT40 + ET) affected strength or ET performance, a 1-way repeated-measures analysis of variance with Bonferroni adjustment was used. To compare the metabolic and mechanical responses induced by each protocol, a 2-way condition (4 levels: ET + RT20 vs ET + RT40 vs RT20 + ET vs RT40 + ET) × time (3 levels: pre vs postendurance vs postresistance) repeated-measures analysis of variance was performed. Bonferroni post hoc tests were used when the interaction was significant. Statistical analyses were performed using SPSS (version 24.0; SPSS Inc, Chicago, IL).

Results

Descriptive Characteristics of the Concurrent Training Protocols

Table 1 shows the physiological and mechanical characteristics obtained throughout the 4 CT protocols. With regard to endurance exercise, the ability to maintain the scheduled relative intensity (90% vVO_2max) was different between protocols ($P = .002$), with the RT40 + ET protocol showing significantly shorter running times than the rest of the protocols. All protocols attained similar peak values in HR, VO_2, respiratory exchange ratio, and VE during the endurance exercise. However, significant differences were observed for peak values of VE/VO_2 ($P = .006$) and VE/VCO_2 ($P = .002$), where the RT40 + ET protocol showed significantly higher VE/VO_2 and VE/VCO_2 values than ET + RT40 and ET + RT20, respectively.

With regard to resistance exercise, the ET + RT40 showed significantly slower Fastest-V than the RT20 + ET and RT40 + ET protocols. However, no significant differences were observed between the repetitions performed during the ET + RT20 and RT20 + ET protocols or between RT40 + ET and ET + RT40. As expected, more repetitions per set were performed during the RT40 + ET and ET + RT40 protocols compared with ET + RT20 and RT20 + ET. Moreover, the Slowest-V and Mean-V were significantly faster for ET + RT20 and RT20 + ET compared with RT40 + ET and ET + RT40.

Table 1 Descriptive Characteristics of Each Concurrent Training Protocol

Physiological and mechanical characteristics	E + R20	E + R40	R20 + E	R40 + E	*P* value protocol effect
Endurance training					
90% vVO_2max, $km·h^{-1}$	14.1 (1.3)	14.1 (1.3)	14.1 (1.3)	14.1 (1.3)	1.00
Running time, s	600 (0)***	600 (0)***	561 (70)***	450 (131)	.002
HR peak, bpm	180.7 (9.3)	181.4 (9.9)	184.1 (11.2)	183.2 (8.4)	.06
Rel-HR peak, %	96.5 (2.9)	96.8 (2.1)	98.2 (2.5)	97.8 (2.1)	.07
VO_2peak, $mL·kg·min^{-1}$	48.5 (4.6)	47.4 (5.6)	48.1 (5.0)	47.7 (7.7)	.90
Rel-VO_2peak, %	90.5 (8.7)	88.3 (9.6)	89.5 (8.1)	88.8 (14.1)	.86
RER peak	1.11 (0.05)	1.11 (0.06)	1.11 (0.06)	1.08 (0.08)	.28
VE peak, $L·min^{-1}$	131.8 (15.8)	129.3 (12.5)	133.9 (14.6)	134.6 (20.9)	.48
VE/VO_2peak	35.1 (3.7)	34.9 (3.2)***	36.0 (3.2)	37.3 (3.9)	.006
VE/VCO_2peak	32.8 (3.1)***	33.7 (3.0)	34.2 (3.0)	36.0 (3.5)	.007
Resistance training					
Load, kg	63.1 (15.4)	61.8 (14.7)	61.5 (16.0)	62.1 (13.8)	.74
Rep/set, n	9.1 (4.6)*,***	14.4 (6.0)	10.0 (4.9)*,***	18.3 (9.0)	.001
Fastest-V, $m·s^{-1}$	0.96 (0.07)	0.94 (0.07) **,***	1.00 (0.02)	0.99 (0.02)	.04
Slowest-V, $m·s^{-1}$	0.76 (0.09) *,***	0.59 (0.09)	0.82 (0.03) *,***	0.56 (0.05)	<.001
Mean-V, $m·s^{-1}$	0.85 (0.08) *,***	0.75 (0.07)	0.88 (0.02) *,***	0.78 (0.04)	<.001
VL, %	20.5 (2.9)*,***	36.1 (5.7)	18.7 (2.4)*,***	36.9 (2.3)	<.001

Abbreviations: 1RM, 1-repetition maximum; bpm, beats per minute; HR, heart rate; RER, respiratory exchange ratio; VL, ventilator equivalent; VO_2max, maximal oxygen uptake. Note: Data are expressed as mean (SD). E + R20: Endurance (10 min at 90% vVO_2max) + Resistance (60% 1RM with 20% VL); E + R40: Endurance (10 min at 90% vVO_2max) + Resistance (60% 1RM with 40% VL); R20 + E: Resistance (60% 1RM with 20% VL) + Endurance (10 min at 90% vVO_2max); R40 + E: Resistance (60% 1RM with 40% VL) + Endurance (10 min at 90% vVO_2max); 90% vVO_2max: 90% of speed associated with the maximum oxygen uptake; running time for the endurance exercise protocol (scheduled time: 600 s); HR peak: peak heart rate attained during the endurance exercise protocol; Rel-HR peak: peak heart rate, relative to maximum heart rate, attained during the endurance exercise protocol; VO_2 peak: peak oxygen uptake attained during the endurance exercise protocol; Rel-VO_2peak: peak oxygen uptake, relative to maximal oxygen uptake, attained during the endurance exercise protocol; RER peak: peak respiratory exchange ratio attained during the endurance exercise protocol; VE peak: peak of ventilation volume attained during the endurance exercise protocol; VE/VO_2peak: peak of ventilatory equivalent for oxygen attained during the endurance exercise protocol; VE/VCO_2 peak: peak of ventilatory equivalent for carbon dioxide attained during the endurance exercise protocol; Load: absolute load employed in the resistance exercise protocol; Rep/set: repetitions performed in each set; Fastest-V: highest velocity measured in the 3 sets; Slowest-V: lowest velocity measured in the 3 sets; Mean-V: mean velocity of all repetitions during the 3 sets; VL: mean percent loss in velocity over the 3 sets; Velocities correspond to the mean concentric propulsive velocity of each repetition.

*Statistically significant differences with E + R40 protocol: $P < .05$. **Statistically significant differences with R20 + E protocol: $P < .05$. ***Statistically significant differences with R40 + E protocol: $P < .05$.

Metabolic and Mechanical Response to the Concurrent Training Protocols

Significant protocol × time interactions were observed for blood lactate concentration ($P < .001$), CMJ height ($P = .006$), and V_1-load ($P = .011$) (Figure 2). The RT20 + ET and RT40 + ET protocols showed higher lactate concentrations and CMJ height impairments than ET + RT20 and ET + RT40 at postendurance ($P < .05$). Moreover, RT40 + ET produced higher impairments in V_1-load than the ET + RT20 and ET + RT40 protocols at postendurance ($P < .05$), without significant differences between the rest of the protocols. At postresistance, RT20 + ET induced significantly lower blood lactate concentration than the rest of the protocols, and the ET + RT20 protocol produced significantly lower blood lactate concentration than ET + RT40. Moreover, the RT40 + ET and ET + RT40 protocols impaired CMJ to a greater degree ($P < .05$) than the RT20 + ET and ET + RT20 protocols at postresistance, without significant differences between protocols with a similar VL magnitude (ET + RT20 vs RT20 + ET and RT40 + ET vs ET + RT40). In addition, ET + RT20 and RT20 + ET showed higher V_1-load values than ET + RT40 at postresistance, and the RT20 + ET protocol produced greater V_1-load performances than the RT40 + ET protocol. No significant differences were observed between RT20 + ET and ET + RT20 for V_1-load at postresistance.

Discussion

This study is the first to examine the effects of altering the exercise sequence and the level of fatigue induced in resistance exercise during concurrent strength and ET. The level of fatigue was standardized via the VL magnitude exerted within the resistance set. Taken together, our results suggest that a high VL magnitude (ie, 40%) during RT resulted in higher metabolic and mechanical stress (ie, greater blood lactate and higher CMJ height and V1-load impairments) and greater residual fatigue on following endurance exercise performance. However, endurance performance was not affected when resistance exercise with a moderate VL was performed before endurance exercise (ie, RT20 + ET). By contrast, the fastest velocity attained during resistance exercise was reduced when endurance exercise was performed before resistance exercise, although no significant differences were observed between the repetitions per set performed for a given VL magnitude, regardless of the exercise sequence (ET + RT20 vs RT20 + ET and RT40 + ET vs ET + RT40). Such findings will enable the design of CT programs to minimize acute detrimental effects on strength and endurance performance.

Running time was impaired after the initial resistance exercise with a high level of fatigue (RT40 + ET), but not when the resistance exercise was performed with moderate levels of fatigue (RT20 + ET). This is in agreement with a previous study,[15] in which performance in a 3-km running time trial was impaired to a greater extent when 10 sets of 10 SQ with an external load corresponding to 80% body mass were performed compared with 5 sets of 10 SQ with the same load. In this regard, previous reports have shown that strength-training sessions with high levels of fatigue within the set (ie, close to failure) resulted in greater impairment of neuromuscular performance and a slower recovery.[29,30] Furthermore, the protocol with high-fatigue RT prior to ET (RT40 + ET) showed increased ventilatory equivalents (VE/VO_2 and VE/VCO_2) compared with those that performed endurance exercise without previous fatigue. These findings indicated that RT

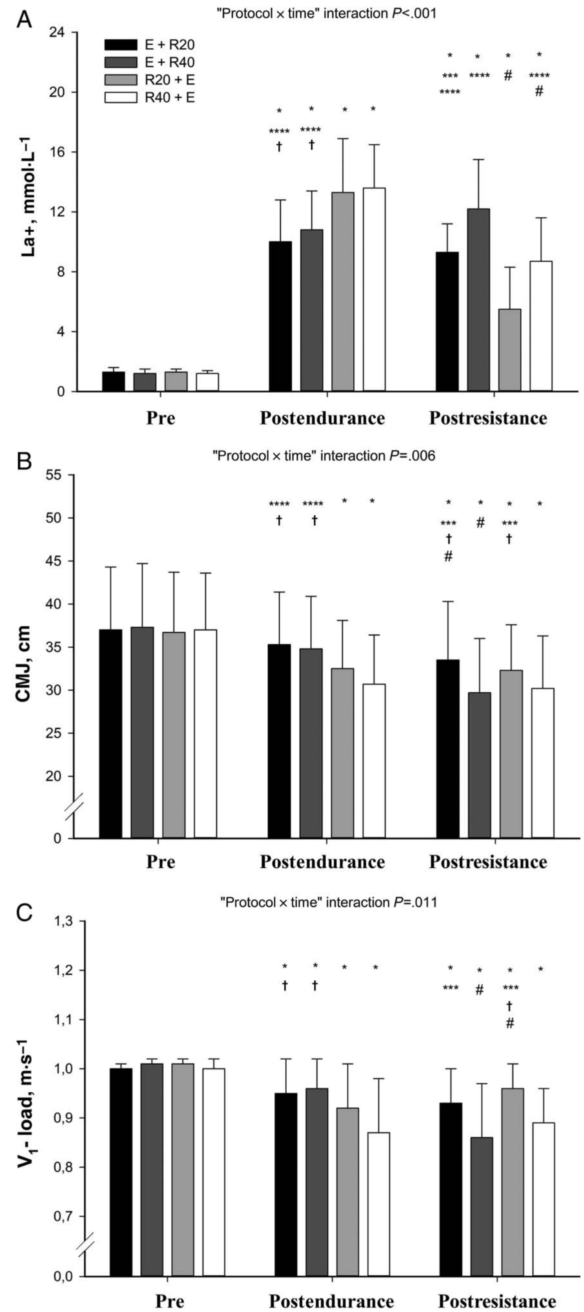

Figure 2 — Metabolic and mechanical response to the concurrent training protocols. Data are expressed as mean (SD). ET + RT20: Endurance (10 min at 90% vVO_2max) + Resistance (60% 1RM with 20% VL); ET + RT40: Endurance (10 min at 90% vVO_2max) + Resistance (60% 1RM with 40% VL); RT20 + ET: Resistance (60% 1RM with 20% VL) + Endurance (10 min at 90% vVO_2max); RT40 + ET: Resistance (60% 1RM with 40% VL) + Endurance (10 min at 90% vVO_2max); Pre: baseline measure; Post-1: measure after the first exercise; Post-2: measure after the second exercise; 1RM indicates 1-repetition maximum; CMJ, countermovement jump; ET, endurance training; La+, blood lactate; RT, resistance training; V_1-load, velocity attained against the absolute load that elicits 1 $m \cdot s^{-1}$ in the baseline measure (Pre); VL, velocity loss; vVO_2max, velocity associated at maximal oxygen uptake. *Statistically significant differences from Pre at the corresponding time point: $P < .05$. **Statistically significant differences from Post-1 at the corresponding time point: $P < .05$. ***Statistically significant differences with ET + RT40 protocol: $P < .05$. ****Statistically significant differences with RT20 + ET protocol: $P < .05$. †Statistically significant differences with RT40 + ET protocol: $P < .05$.

with high levels of fatigue increased the physiological cost of endurance running exercise, which is consistent with previous research that reported increased cardiorespiratory parameters during submaximal running after strength training,[11,12,31] accompanied by reductions in muscle strength following the resistance–endurance sequence.[31,32] Impaired muscle strength may require that a greater number of motor units be recruited to maintain running intensity, resulting in disturbances in the metabolic system.[33] Indeed, Burt et al[15] observed an increased neuromuscular activity and a higher blood lactate concentration during endurance running exercise following resistance exercise, which was attributed to a compensatory increase in motor unit recruitment. Interestingly, our findings showed higher reductions in strength markers (CMJ height and V1-load), along with elevated blood lactate concentration, following endurance exercise when resistance exercise was previously performed (RT + ET), when high levels of fatigue within the strength exercise were induced. These findings suggest that high-fatigue resistance exercise before endurance exercise (RT40 + ET) should be avoided in order to prevent decreased quality in the subsequent ET session (ie, impairments in running time and increased physiological cost).

Conducting endurance exercise prior to strength training resulted in impaired strength training performance (ie, ET + RT40 showed a significantly slower Fastest-V than the RT20 + ET and RT40 + ET protocols), although no significant differences were observed in the repetitions per set performed for a given VL threshold, regardless of the exercise sequence (ET + RT20: 10.0 [4.9] vs RT20 + ET: 9.1 [4.6]; and RT40 + ET: 18.3 [9.0] vs ET + RT40: 14.4 [6.0]). These findings indicate that when athletes performed endurance exercise prior to strength training they were more fatigued during the first repetitions, resulting in a slower Fastest-V. However, as the athletes went on to perform further repetitions, fatigue developed throughout the set, and the differences in velocity between protocols became less pronounced, resulting in no differences in the mean velocity of all repetitions (Mean-V) and in the last repetition completed in each protocol (Slowest-V). In addition, no significant differences were found between protocols with a similar VL threshold for V_1-load and CMJ height after completing the resistance exercise (ie, at post-resistance), regardless of the exercise sequence. Contrary to our results, previous studies have observed a reduced number of repetitions that could be performed to failure during strength training after an endurance exercise.[8,9] Conversely, de Souza et al[34] found no difference in the ability to complete repetitions after an aerobic exercise protocol. The different training intensities, volumes, and exercises employed may hamper a direct comparison between studies. Blood lactate concentrations were greater when ET was conducted prior to strength training and when a high VL was attained within RT. It is worth noticing that the protocols that incurred high VL levels within the RT set (ie, 40%) induced the highest CMJ and V_1-load impairments, regardless of the exercise sequence (Figure 2). In accordance with these results, a previous study also observed that performing the SQ exercise with 60% 1RM until attaining a 40% VL resulted in greater fatigue and a slower rate of recovery, measured by running sprint, CMJ, and V_1-load, than a lower VL threshold (20%) with the same relative load (60% 1RM).[35] Furthermore, previous studies have reported that lower VL thresholds (10%–20% VL) produced similar or even superior strength gains to higher VL magnitudes (30%–40%).[18,27] Therefore, in agreement with previous evidence, both endurance and resistance exercise may impair the performance of following exercise,[11,12,31] although this is the first study to show that setting a moderate VL threshold (20% VL) in the RT set during CT may avoid performing fatiguing repetitions, which may compromise the quality of the following endurance sessions.

Practical Applications

Our findings further demonstrate the need for strategic training prescriptions during CT approaches to minimize the transfer of fatigue between endurance and strength training sessions. With regard to exercise sequence, the most relevant quality for an athlete's discipline should be trained first in order to prevent the residual fatigue evoked by the first exercise. With regard to the VL threshold incurred during RT, the extent to which ET performance is compromised appears to be dependent on the level of fatigue induced during resistance exercise, since lower-fatigue resistance exercise did not induce significant impairments in performance in the following endurance exercise (ie, RT20 + ET), while higher-fatigue resistance exercise resulted in detrimental effects on endurance exercise performance (ie, RT40 + ET). Therefore, high-fatigue resistance exercise before endurance exercise should be avoided.

Conclusions

The results of this study indicate that the residual fatigue induced by the first exercise impairs the quality of the subsequent exercise during training. The present study's data also show that a high VL magnitude (40% VL) within the RT session induced higher metabolic and mechanical stress, as well as greater residual fatigue on the following ET performance.

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